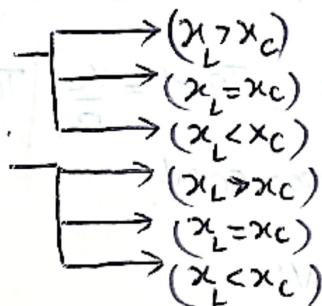


ALTERNATING CURRENT

SUB-TOPICS

- ⇒ i_{avg} i_{rms}
- ⇒ V_{avg} V_{rms}
- ⇒ Resistor with AC.
- ⇒ Resistor Capacitor with AC
- ⇒ Capacitor with AC.
- ⇒ Inductor with AC
- ⇒ Phasors.
- ⇒ L-R Series with AC
- ⇒ LC Series with AC
- ⇒ LCR Series with AC.
- ⇒ Powers Supplied by AC Source.
- ⇒ Avg Power in one Complete Cycle.
- ⇒ Sharpness of Resonance.
- ⇒ Q-factor.
- ⇒ Wattless Current.
- ⇒ Choke coil
- ⇒ Transformers
- ⇒ Parallel Circuits



where,

$L \rightarrow$ Self Inductance.

$\omega \rightarrow \frac{2\pi}{T}$ $2\pi f$

i. T: Time Period.

ii. f: frequency

Units:

Hz...

Identify by Units..

$$X_L = \omega L$$

$$X_C = \frac{1}{\omega C}$$

CONCEPTS & KEY POINTS...

$$V_{rms} = \sqrt{\frac{\int V^2 dt}{\int dt}}$$

$$P = VI$$

$$1. i_{avg} = \frac{\int i dt}{\int dt}$$

$$2. i_{rms} = \sqrt{\frac{\int i^2 dt}{\int dt}}$$

- Avg Value of $\sin^2 x$ is $\frac{1}{2}$
- Avg Value of Constant: Constant
- Avg Value of $\sin x$: 0
- Avg Value of $\cos x$: 0

$$I_{rms} = \frac{V_{rms}}{Z} \quad I_{rms} = \frac{I_{max}}{\sqrt{2}} \quad I_{rms} = \frac{V_{rms}}{X_C}$$

$$Z = \sqrt{R^2 + X_C^2} \Rightarrow Z = \sqrt{R^2 + (X_C - X_L)^2} \quad ***$$

$$\tan \phi = \frac{X_C}{R}, \quad \tan \phi = \frac{X_L}{R} \quad \boxed{\tan \phi = \frac{X_L - X_C}{R}} \quad ***$$

$$\text{Power factor} = \cos \phi$$

$$P = VI$$

$$P = V_{max} \sin \omega t \cdot I_{max} \cos \omega t = \frac{1}{2} V_{max} I_{max} \sin 2\omega t$$

NOTE: Pure capacitive circuit Avg Power is zero...

$$\frac{dq}{dt} = V_{max} C \omega \cos \omega t$$

$$i = \frac{V}{\sqrt{R^2 + X_L^2}} \quad i = \frac{V_0}{\sqrt{R^2 + (X_C - X_L)^2}} \quad (\text{Amplitude})$$

Phase Diff B/w Primary/secondary Voltage is π . ***

Step up : $N_s > N_p \Rightarrow E_s > E_p$

Step Down : $N_s < N_p \Rightarrow E_s < E_p$

One complete cycle Avg Current Change is always Zero.

$$\boxed{Q = \frac{1}{R} \sqrt{\frac{L}{C}}}$$

Choke coil: Current Lags behind emf by $\pi/2$.

Transformers:

Changes an Alternating Potential difference from One Value to another of Greater or Smaller Value.

Two coils $\swarrow$ Primary
 $\searrow$ Secondary ...

$$\frac{1}{\omega} = \frac{1}{\omega}$$

$$\frac{E_{\text{second}}}{E_{\text{Primary}}} = \frac{\text{No. of Turns of Second}}{\text{No. of Turns of Primary}} = \frac{I_{\text{Primary}}}{I_{\text{Second}}}$$

Average Power is Power dissipated by resistor ...

$$P = (i_{\text{rms}})^2 R$$

$$P = V_{\text{rms}} i_{\text{rms}} \cos \phi$$

: Power factor $= \cos \phi = \frac{R}{Z}$. Q-factor $= \frac{X_L}{R} = \frac{X_C}{R}$

: Phase Difference is always 180° (or 90°) when Resistance and Inductance are joined in Series ...

: Phase Difference is always Zero for the Case of Resonance. $\boxed{Z = R}$

$$\omega = \frac{1}{\sqrt{LC}}$$

$$f = \frac{1}{2\pi\sqrt{LC}}$$

$$\boxed{X_C = \frac{1}{\omega C}} \quad X_C \propto \frac{1}{\omega} \quad X_L = \omega L$$

$$\boxed{V = L \cdot \frac{di}{dt}}$$

Energy Stored in coil in ts is $\frac{1}{2} Li^2$

$$\frac{Q_1}{Q_2} = \frac{N_1}{N_2} = \frac{i_2}{i_1}$$

$$i = \frac{V_0}{X_c}$$

$$X_c = \frac{1}{\omega C}$$

where $C = \frac{\epsilon_0 \epsilon_r A}{d}$

$$\epsilon_0 \rightarrow 9 \times 10^9$$

$$A \rightarrow \text{Area}$$

$$d \rightarrow \text{Sep b/w plates}$$